

Figure 4D: Control-Layer Contribution

Control layer	Primary benefit	What moves	SLA Δ (pp of S0)	OPEX Δ (pp of S0)
PL	Optimal placement (where)	OPEX, E2E latency	−79.3	−41.3
PM	Elastic compute (how much)	CPU/GPU util., queue backlog	−13.5	−10.4
PS	Channel adaptation (how radio)	SLA violation, throughput	−6.9	−0.0
PL + PM + PS	End-to-end multi-timescale	SLA-OPEX Pareto frontier	−99.8 total	−51.7 total